

PRIYANKA SHARMA

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EDUCATION

National Institute of Technology, Agartala

2024 – 2028

B.Tech in Chemical Engineering

- Relevant Coursework: Data Structures & Algorithms, Chemical Engineering Principles
- Additional Learning: Machine Learning, Deep Learning, Data Science, Graph Neural Networks

SKILLS

Programming: Python, C++

Machine Learning: Logistic Regression, Random Forest, XGBoost, Scikit-learn

Deep Learning: PyTorch, Graph Neural Networks (GNNs), TensorFlow/Keras

Libraries: NumPy, Pandas, Matplotlib, RDKit

Tools: Git, Google Colab, Jupyter Notebook

PROJECTS

Molecular Solubility Prediction 

2025

- Built machine learning models to predict aqueous solubility using the ESOL dataset.
- Implemented Linear Regression, Random Forest, and XGBoost models.
- Engineered molecular fingerprints using RDKit and achieved an R^2 score of 0.89.
- Tools: Python, Pandas, Scikit-learn, RDKit.

Toxicity Prediction using Graph Neural Networks 

2025

- Developed a multi-task molecular toxicity prediction model using the Tox21 dataset.
- Implemented Graph Neural Networks with PyTorch Geometric to learn from molecular graphs.
- Evaluated performance using ROC-AUC and compared with traditional ML baselines.
- Tools: Python, PyTorch, PyTorch Geometric, RDKit.